

Smart Exam Hall Anti-Cheating Seat Allocation System

Documentation

1. System Description

The Smart Exam Hall Anti-Cheating Seat Allocation System is a web-based application designed to automate seat allocation and reduce cheating possibilities during examinations.

The system uses randomized allocation and constraint checking to prevent students from the same department from sitting adjacent to one another.

2. Functional Requirements

Input

- Department information
- Hall dimensions

Output

- Final seat allocation
- Risk visualization
- Hover-based seat details

Functionalities

- Allocate seats
 - Shuffle seating
 - Reset hall
 - Highlight risky neighbors
 - Display student information
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3. Non-Functional Requirements

- Fast execution
- Responsive interface
- Low memory usage

- Easy visualization
 - Scalable architecture
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4. Objectives

- Automate exam seat allocation
 - Reduce cheating possibilities
 - Separate same-department students
 - Improve secure and fair seat allocation
 - Improve fairness and transparency
 - Provide interactive visualization for monitoring
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5. Technologies Used

Technology Purpose

HTML Structure of the application

CSS User interface styling

JavaScript Logic and algorithms

6. System Modules

6.1 User Interface Module

This module manages the graphical layout of the system.

Features

- Interactive seat grid
 - Allocate, Shuffle, and Reset buttons
 - Hover-based student information panel
 - Risk highlighting
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6.2 Allocation Module

This module handles seat allocation.

Functions

- Randomizes seat order
- Checks seating constraints
- Places students into seats safely

Constraints Used

- Same department students cannot sit adjacent to each other.
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6.3 Risk Analysis Module

This module calculates and visualizes cheating risk.

Risk Factors

- Same department adjacency
- Neighbor relationships

Risk Levels

Level	Color
Low	Green
Medium	Yellow
High	Red

7. Algorithms Used

7.1 Fisher–Yates Shuffle Algorithm

The Fisher–Yates Shuffle Algorithm is used to randomize student order before seat allocation.

Working Principle

1. Start from the last element.
2. Select a random index.
3. Swap elements.
4. Repeat until shuffled.

Advantages

- Fair randomization
 - Efficient execution
 - $O(n)$ time complexity
 - $O(1)$ space complexity
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7.2 Constraint-Based Allocation

Before placing a student, the algorithm checks neighboring seats.

Directions Checked

$[(0,-1), (-1,0)]$

Meaning:

- Left neighbor
- Top neighbor

Why only these directions?

The hall is filled row-wise from left to right. Therefore, only previously allocated seats need to be checked.

Complexity

- Time Complexity: $O(1)$
 - Space Complexity: $O(1)$
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8. Working Procedure

Step 1

Initialize the exam hall grid.

Step 2

Randomize student order using Fisher–Yates Shuffle.

Step 3

Traverse the hall row by row.

Step 4

Check if the current seat is safe using the constraint function.

Step 5

If constraints are satisfied, place the student.

Step 6

Calculate seat risk.

Step 7

Render the final seating arrangement visually.

9. User Interface Features

9.1 Hover Information Panel

The system displays basic student information when the user hovers over a seat.

Information Displayed

- Student ID
- Department

This improves monitoring and helps invigilators identify students quickly.

9.2 Neighbor Expansion Effect

When hovering over a seat, neighboring seats with copying possibility expand visually.

This helps invigilators identify high-risk regions quickly.

9.3 Risk Visualization

Seats are visually highlighted according to their risk level.

- Green → Low Risk
 - Yellow → Medium Risk
 - Red → High Risk
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10. Advantages

- Automated seat allocation

- Reduced manual effort
 - Improved fairness
 - Reduced cheating probability
 - Interactive visualization
 - Efficient execution
 - Scalable architecture
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11. Limitations

- Rule-based system
 - Limited neighbor analysis
 - No database integration
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12. Future Enhancements

- Advanced seat optimization
 - Database connectivity
 - Multi-hall support
 - Real-time monitoring
 - Facial recognition integration
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13. Conclusion

The Smart Exam Hall Anti-Cheating Seat Allocation System successfully automates exam seating using randomized allocation and constraint-based validation. The system improves fairness, reduces same-department adjacency, and provides interactive visualization features for effective monitoring.

The project demonstrates the practical use of algorithms and visualization techniques in examination management systems.

14. References

1. Fisher–Yates Shuffle Algorithm

2. JavaScript Documentation – MDN
3. HTML and CSS Documentation
4. Data Structures and Algorithms Concepts